

Spreading 46 Points in the Plane

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Abstract

We place 46 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 43.71614+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 47-point configuration gives 44.59453; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 46 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 43.71614\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 46$ is not on the page. Deleting a point from our 47-point configuration gives 44.59453; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n46.txt
```

in <https://github.com/Ge-limin/math719>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
43716145205106803030588353
denominator:
999999999998347352488213
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

46 points, x and y per line. File: friedman-maxmin2d/coords/n46.txt.

```
-3.901327822815 7.624274337806
0.641328071575 9.016528165956
1.589782207730 8.699613742790
1.355038123193 7.727556535227
-2.277536614956 11.624312401137
-2.306868760822 8.764831545802
-1.279416245417 11.685596395067
-2.370384808034 5.164405338175
-1.577726011931 6.053579626689
-4.769110008069 10.219735275097
-2.573672772577 6.143524331696
-0.130776005953 10.095911215973
-4.970093790216 7.872466780832
-1.754735080464 7.931075946712
0.406583987038 8.044470958394
-0.616223032484 6.328374132666
-0.412935067941 5.349255139146
-1.308748391283 8.826115539733
-4.246585356189 8.562782321649
-3.253753895413 8.443259560252
-3.233081245367 11.329465899901
0.333362871504 6.014867177484
0.058979840523 11.161118991038
-1.086320636365 9.801064714737
-0.863892881446 10.776013889741
-0.353203760871 9.120962040969
-3.039985636315 9.444934219571
1.390011080940 9.679456330608
-3.382873097455 6.731057168410
-2.807005317660 7.548600011443
-0.039441370420 7.148988386775
-1.997804992782 6.961067174728
-4.035434090187 9.540235718423
-4.129171036899 6.065445130072
-1.374438047388 5.074460633169
-1.035633741144 7.236170727428
-1.862013250985 10.714729895811
-2.084441005903 9.739780720807
-4.624836256943 6.933958796999
-0.589647051963 8.131210320448
-5.001523284678 9.220033870786
-3.319970712021 5.477912293358
0.909145097090 6.832470277221
0.792096716016 10.481016317270
-2.817557881397 10.419883394575
-3.813808171189 10.515367479791
```